



EE2012

Monkey Don't Fall into the Same Hole Twice.

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Introduction

Probability is actually just about managing a chaotic zoo of monkeys (events) and bananas (outcomes).

- **The Banana Pile (Sample Space):** Imagine every possible thing that could happen is a banana in a giant pile. The size of this entire pile is **1** (or 100%).
- **Events:** An "Event" is just a specific group of bananas. "Rolling a 6" is a small bunch of bananas. "Rolling an even number" is a larger bunch.
- **Probability:** Measuring the weight of a specific bunch compared to the whole pile.
- **Random Variable (RV):** This isn't algebra x . This is a **Monkey**. We don't know exactly what banana the monkey will grab, but we know his habits (distribution).

1 Tutorial 1: The Basics (Counting Monkeys)

The most fundamental skill in probability is counting. If you can count the "Winning Outcomes" and divide by the "Total Outcomes", you have the probability.

Concept: The Venn Diagram Rules

Concept Guide: Inclusion-Exclusion Principle

The Problem: Sometimes a monkey joins two clubs (e.g., the Basketball Club AND the Soccer Club). **The Analogy:** If you count all the Basketball players, you count the dual-athletes once. If you count all the Soccer players, you count the dual-athletes *again*. If you just add the two lists together, you have counted the dual-athletes twice! You must subtract them once to correct the total.

$$\text{Total} = \text{Group A} + \text{Group B} - (\text{Overlap}) + \text{Neither}$$

Q1: The Venn Diagram Monkeys

Q1: Basketball vs. Soccer

The Setup: Total = 10 monkeys.

- 6 like Red Bananas (Basketball).
- 7 like Yellow Bananas (Soccer).
- 1 monkey is grumpy (Neither).

(a) Find the probability a student plays BOTH. *Step-by-Step Logic:*

1. First, remove the grumpy monkey. $10 - 1 = 9$ active players.
2. Now add the rosters: 6 (Basket) + 7 (Soccer) = 13 spots on the rosters.
3. But we only have 9 active players! Why are there 13 spots?
4. Because $13 - 9 = 4$ monkeys signed up for both teams.
5. **Answer:** $4/10 = 0.4$.

(b) Pick two students (A then B). Probability A is Basketball, B is Soccer? *Concept: Sampling WITHOUT Replacement.* Imagine reaching into the cage.

- **Step 1 (Monkey A):** We need a Basketball player. There are 6.
- **Step 2 (Monkey B):** We need a Soccer player. Monkey A is gone (he's outside eating a banana). There are only 9 monkeys left inside.
- **The Catch:** Does Monkey A play soccer too? If he does, there are fewer soccer players left for Step 2. We must split this into scenarios.

Scenario 1: Monkey A plays ONLY Basketball.

- How many play only BB? Total BB (6) - Both (4) = 2.
- $\text{Prob}(\text{Pick Only BB}) = 2/10$.
- Now 9 monkeys left. All 7 soccer players are still inside.
- $\text{Prob}(\text{Pick Soccer}) = 7/9$.
- Result: $\frac{2}{10} \times \frac{7}{9} = \frac{14}{90}$.

Scenario 2: Monkey A plays BOTH.

- There are 4 such monkeys. $\text{Prob}(\text{Pick Both}) = 4/10$.
- Now 9 monkeys left. One soccer player (Monkey A) is gone. Only 6 soccer players remain.
- $\text{Prob}(\text{Pick Soccer}) = 6/9$.
- Result: $\frac{4}{10} \times \frac{6}{9} = \frac{24}{90}$.

Final Answer: Add the scenarios: $\frac{14}{90} + \frac{24}{90} = \frac{38}{90} = \frac{19}{45}$.

Concept: Independence

Concept Guide: Independence

Two events are **Independent** if the first one happening doesn't change the probability of the second one.

- **Example:** Rolling a 6 on dice #1 doesn't change the luck of dice #2.
- **Rule:** If A and B are independent, $P(A \text{ and } B) = P(A) \times P(B)$.
- *Analogy:* If two monkeys are in separate soundproof cages, Monkey A eating a banana doesn't make Monkey B hungry. You just multiply their probabilities.

Q2: Dice Rolls

Q2: The Sequence 1-2-3-4-5-6

(a) **Fair Dice Logic:** We need specific numbers in a specific order.

- Roll 1 must be a '1' (1/6 chance).
- Roll 2 must be a '2' (1/6 chance).
- ...
- Since dice have no memory, multiply them: $(\frac{1}{6})^6 = \frac{1}{46656}$.

(b) **Loaded Dice (The Cheating Monkey) Setup:** Even numbers are twice as likely as odd.

- Let p be the probability of an Odd number. Then $2p$ is the probability of an Even number.
- The sum must be 1: $3(\text{Odds}) + 3(\text{Evens}) = 1$.
- $3(p) + 3(2p) = 1 \Rightarrow 9p = 1 \Rightarrow p = 1/9$.
- So, $P(\text{Odd}) = 1/9$, $P(\text{Even}) = 2/9$.
- Sequence 1-2-3-4-5-6 is Odd-Even-Odd-Even-Odd-Even.
- Calc: $\frac{1}{9} \times \frac{2}{9} \times \frac{1}{9} \times \frac{2}{9} \times \frac{1}{9} \times \frac{2}{9} = \frac{8}{9^6}$.

Concept: The Lazy Monkey Trick (Complements)

Concept Guide: The Complement Rule

Sometimes, counting what you *want* is hard (e.g., "At least one banana"). It is much easier to count what you *don't* want ("Zero bananas") and subtract from the total pile (1).

$$P(\text{At least one}) = 1 - P(\text{None})$$

Q3: Combinations

Q3: Dice Strategy

(a) **10 rolls, at least one "6"** *Direct way (Hard):* Calculate prob of one 6, plus prob of two 6s, plus three 6s... *Lazy Monkey way:* Calculate prob of ZERO 6s.

- Chance of NOT rolling a 6 once: $5/6$.
- Chance of NOT rolling a 6 ten times: $(5/6)^{10}$.
- Answer: $1 - (5/6)^{10}$.

(b) **5 rolls: Two 1s, Two 3s, One 6** *Logic: Multinomials are about assigning seats on a bus.*

- We have 5 seats (Dice 1 to 5).
- "Who wants to be a 1?" Choose 2 seats: $\binom{5}{2} = 10$ ways.
- "Who wants to be a 3?" From the remaining 3 seats, choose 2: $\binom{3}{2} = 3$ ways.
- "Who wants to be a 6?" The last seat is taken: 1 way.
- Total Arrangements: $10 \times 3 \times 1 = 30$.
- Total Possible Rolls: 6^5 .
- Answer: $\frac{30}{6^5}$.

(c) **Negative Binomial (The Waiting Game)** *Question: Probability the k -th success happens exactly on roll n ?* *Logic:* This is a strict story.

- Act 1: In the first $n - 1$ rolls, we must have gotten exactly $k - 1$ successes. (Use Binomial Formula).
- Act 2: The very last roll (n) MUST be a success. (Multiply by p).
- Formula:
$$\underbrace{\binom{n-1}{k-1} p^{k-1} (1-p)^{(n-1)-(k-1)}}_{\text{Act 1}} \times \underbrace{p}_{\text{Act 2}}.$$

Q6 & Q7: The Race Condition

Q6 & Q7: Who wins first?

Dice: Sum 5 happens BEFORE Sum 7 *Concept: Relative Probability.* Imagine a race. We ignore every result that isn't a 5 or a 7. They don't count.

- Ways to make 5: (1,4), (4,1), (2,3), (3,2). That's 4 horses for Team 5.
- Ways to make 7: (1,6), (6,1), (2,5), (5,2), (3,4), (4,3). That's 6 horses for Team 7.
- Total horses in the race: $4 + 6 = 10$.
- Chance Team 5 wins: $4/10 = 2/5$.

2 Tutorial 2: Conditional Probability (The Fence)

Concept: The Fence

Concept Guide: Conditional Probability

The notation $P(A|B)$ means "Probability of A **given** B". **The Analogy:** Imagine the Sample Space is a big field of monkeys. When we say "Given B", we build a **Fence** around only the monkeys inside event B. We kick everyone else out. Now, inside this fence, we calculate the fraction of monkeys who are also A.

$$P(A|B) = \frac{\text{Monkeys in A AND B}}{\text{Total Monkeys in Fence B}}$$

Q1: Dice Conditions

Q1: Conditional Dice

(a) Find $P(A|B)$ where $B = \{\text{min} = 2\}$. *Step 1: Build the Fence (Event B).* We only care about rolls where the *smallest* number is 2. The Fence contains: (2,2), (2,3), (2,4), (2,5), (2,6) AND (3,2), (4,2), (5,2), (6,2). Total outcomes in fence = 9.

Step 2: Look inside the Fence for Event A (Max=m).

- **Case $m = 1$:** Can the max be 1 if the min is 2? Impossible. Prob = 0.
- **Case $m = 2$:** We need Max=2 AND Min=2. Inside our fence, only (2,2) fits. That's 1 outcome. Prob = 1/9.
- **Case $m = 5$:** We need Max=5 AND Min=2. Inside our fence, (2,5) and (5,2) fit. That's 2 outcomes. Prob = 2/9.

(b) $P(X < Y | Y \geq 2)$ *Step 1: Build the Fence ($Y \geq 2$).* Ignore any roll where the second die is 1. We keep rows 2, 3, 4, 5, 6. Total in fence = 30 outcomes (36 total - 6 ones). *Step 2: Count Target ($X < Y$).* Inside the fence, how many have first die smaller than second? (1,2), (1,3)... count them up. There are 15. *Result:* $15/30 = 1/2$.

Concept: Bayes' Theorem (The Detective)

Concept Guide: Bayes' Theorem

Bayes' Theorem is used for "Reverse Probability". **Forward:** If I have a biased coin, it's likely to land Heads. **Reverse (Bayes):** I saw Heads. Is it likely I have the biased coin?

$$P(\text{Cause}|\text{Effect}) = \frac{P(\text{Effect}|\text{Cause}) \times P(\text{Cause})}{P(\text{Total Effect})}$$

Q2: The Mystery Coins

Q2: Who dropped the banana?

The Setup: Coin A is Biased ($3/4$ Heads). Coin B is Fair ($1/2$ Heads). You pick one at random (50/50 chance). You flip HEADS. **Question:** What is the chance you are holding Coin A?

The Detective Work:

1. **Evidence:** We got Heads.
2. **Suspects:** Coin A and Coin B.
3. **Likelihood:** How good is Coin A at producing Heads? $3/4$. How good is Coin B? $1/2$.
4. **Total Probability of Evidence:** If we ran this experiment 100 times...
 - 50 times we hold Coin A $\rightarrow$ produces 37.5 Heads.
 - 50 times we hold Coin B $\rightarrow$ produces 25 Heads.
 - Total Heads = 62.5.
5. **Conclusion:** What fraction of those 62.5 Heads came from Coin A?

$$\frac{37.5}{62.5} = \frac{3}{5}$$

Formulaic approach: Denominator (Total Prob): $(0.5 \times 0.75) + (0.5 \times 0.5) = 0.625$. Numerator (Coin A): $0.5 \times 0.75 = 0.375$. Result: $0.375/0.625 = 3/5$.

Q4: False Positives

Q4: Virus Test

Is the kit useful? This is a classic Bayes trap. **Setup:** Disease is rare (10^{-4}). False Positive rate is high (10%). **Intuition:** Imagine 10,000 monkeys.

- **Sick Monkeys:** Only 1 is sick. He tests Positive. (1 True Positive).
- **Healthy Monkeys:** 9,999 are healthy. But 10% test Positive by mistake. That's 999 monkeys! (999 False Positives).

The Result: If the alarm rings (Positive Test), you are one of 1,000 monkeys (1 Real + 999 False). Probability you are actually sick: $1/1000 = 0.1\%$. **Conclusion:** The test is useless because the False Positive rate overwhelms the actual infection rate.

3 Tutorial 3: Random Variables (Banana Recipes)

Concept: Discrete RVs

Concept Guide: The Recipe (PMF) and Balance (Expectation)

We stop dealing with "Heads/Tails" and start using numbers.

- **PMF (Probability Mass Function):** A recipe card. "To make this distribution, take 10% of zero, 20% of one..."
- **Expectation ($E[X]$):** The Center of Mass. If you put weights on a seesaw at positions 0, 1, 2... where would the seesaw balance?

$$E[X] = \sum x \cdot P(x)$$

Q2: Expectation of Functions

Q2: Wi-Fi Packets

(c) **Probability Message Correct** *The Setup:* A file is split into N packets. N can be 1, 2, or 3 (with different probs). Every single packet must survive for the message to arrive. *Logic:* Total Probability Law. We solve the problem for each specific case of N , then average them.

- **Case $N=1$:** Needs 1 packet to survive. Prob = p . (Happens 6/11 of the time).
- **Case $N=2$:** Needs 2 packets to survive. Prob = $p \times p = p^2$. (Happens 3/11 of the time).
- **Case $N=3$:** Needs 3 packets. Prob = p^3 . (Happens 2/11 of the time).

Combine: Total Success = $\frac{6}{11}p + \frac{3}{11}p^2 + \frac{2}{11}p^3$.

Q4: Poisson Distribution

Q4: The Bus Stop

The Poisson Distribution is the "Law of Rare Events". It counts things happening over time (buses, emails, raindrops). **Parameter λ (Lambda):** The rate. "How many buses usually come?"

$$P(k \text{ events}) = \frac{e^{-\lambda} \lambda^k}{k!}$$

(d) **99% chance of at least 1 bus** *Lazy Monkey Strategy:* "At least one" is the opposite of "None". We want $1 - P(\text{None}) = 0.99$. So $P(\text{None}) = 0.01$. For Poisson, $P(0) = e^{-\lambda}$.

- $e^{-\lambda} = 0.01$.
- $-\lambda = \ln(0.01) \approx -4.6$.
- $\lambda = 4.6$.
- Since $\lambda = T/5$, then $T = 5 \times 4.6 = 23$ minutes.

Q6: LOTUS Rule

Concept Guide: LOTUS

Law Of The Unconscious Statistician. This theorem saves you work. If you want to find the average of a function $g(X)$, you **don't** need to figure out the complex probability distribution of $g(X)$. You can just use the original probabilities of X .

$$E[g(X)] = \sum g(x) \cdot P(X = x)$$

Analogy: If you want to know the average weight of squared bananas, just take each banana, square its weight, and multiply by its probability. You don't need to find the "distribution of squared bananas" first.

Q6: The Hard Math

Find $E[(1-p)^X]$ for Binomial Step-by-Step:

1. Apply LOTUS: Sum of $(1-p)^x \times P(X = x)$.
2. $P(X = x)$ is the Binomial formula: $\binom{n}{x} p^x (1-p)^{n-x}$.
3. Combine terms: We have $(1-p)^x$ and $(1-p)^{n-x}$. Wait, that doesn't combine well.
4. Actually, group the terms with power x : We have $(1-p)^x$ and p^x . Combine them to $((1-p)p)^x$.
5. Regroup: $\sum \binom{n}{x} [(1-p)p]^x (1-p)^{n-x}$.
6. Recognize the Binomial Expansion Pattern: $(A+B)^n = \sum \binom{n}{x} A^x B^{n-x}$.
7. Result: $((1-p)p + (1-p))^n$. This simplifies to $(1-p^2)^n$.

4 Tutorial 4: Multiple Monkeys & Transformations

Q1: Joint Probabilities

Q1: Independent X and Y

(a) **Find $P(X \geq Y)$** *Method: Brute Force the winning pairs.* We are comparing two monkeys. Since they are independent, we just list every scenario where Monkey X wins or ties, calculate the probability, and add them up.

- **Scenario $X=1$:** Probability $1/3$. X is holding a 1. Y holds -1, 0, or 1. X wins or ties all of them. (Contribution: $1/3 \times 1$).
- **Scenario $X=0$:** Probability $1/3$. X is holding a 0. X wins if Y is -1 or 0. Y's prob for this is $0.5 + 0.2 = 0.7$. (Contribution: $1/3 \times 0.7$).
- **Scenario $X=-1$:** Probability $1/3$. X only wins if Y is -1. Prob 0.5. (Contribution: $1/3 \times 0.5$).

(b) **The Square Machine (X^2, Y^2)** *Concept: Mapping Functions.* Imagine a machine that takes a number and squares it. If Monkey X holds -1, machine outputs 1. If Monkey X holds 1, machine outputs 1. The output "1" is now twice as likely because two inputs map to it. $P(X^2 = 1) = P(X = -1) + P(X = 1)$. Independence is preserved because the machines for X and Y are separate.

Q5: Linearity of Expectation

Concept Guide: Linearity of Expectation

This is a superpower.

$$E[A + B] = E[A] + E[B]$$

Why it's cool: It works **even if A and B are dependent**. It doesn't matter if Monkey A and Monkey B are handcuffed together. The average bananas eaten by the group is always the sum of the average bananas eaten by each monkey.

Q5: Coin Game Profit

Profit $Y = X^2 + X - 4$ We need the average profit: $E[Y]$. Using Linearity: $E[Y] = E[X^2] + E[X] - 4$.

- $E[X]$ is easy. It's a Binomial (4 flips, fair). Average heads = 2.
- $E[X^2]$ is harder. We use the variance formula: $Var(X) = E[X^2] - (E[X])^2$.
- Rearrange: $E[X^2] = Var(X) + (E[X])^2$.
- For Binomial, $Var = np(1 - p) = 4(0.5)(0.5) = 1$.
- So $E[X^2] = 1 + 2^2 = 5$.
- Result: $5 + 2 - 4 = 3$.

Q8: The Geometric Time

Q8: Transmission

The Setup: You transmit a packet. It fails (NACK) or succeeds (ACK). If it fails, you retry. This is a **Geometric Distribution** (counting trials until success). **The Twist (Time):** Each trial takes 1ms to send + 1ms to reply = 2ms. *Except* the last one. Once the receiver gets the good packet, the process is technically "done" for him immediately (1ms), even though the ACK is still flying back. **Mapping:** If X is number of tries:

- 1 try: 1ms.
- 2 tries: 2ms (fail) + 1ms (success) = 3ms.
- 3 tries: 2ms + 2ms + 1ms = 5ms.

Formula: $T = 2X - 1$. To find the probability of time t , just ask "What x gives me this t ?" and use the prob of that x .

5 Tutorial 5: Continuous Variables (Measuring Bananas)

Concept: PDF vs CDF

Concept Guide: The Banana Pile Shape

- **PDF ($f(x)$):** The **Density**. Imagine piling sand to make a shape. The height of the sand is the PDF. The higher the hill, the more likely the value is there.
- **CDF ($F(x)$):** The **Sum**. Imagine walking from left to right with a shovel, scooping up the sand. The CDF is "how much sand do I have in my bucket so far?". It starts at 0 and ends at 1 (full bucket).
- **Relationship:**
 - Integrate Density (PDF) $\rightarrow$ Total Weight (CDF).
 - Differentiate Weight (CDF) $\rightarrow$ Density (PDF).

Q1: Exponential Distribution

Q1: The Memoryless Banana

(c) **Memoryless Property** $P(X > 4 | X > 2) = P(X > 2)$. *Analogy:* Imagine waiting for a bus that arrives completely randomly. You have been waiting 20 minutes. You ask: "What is the chance I have to wait 10 *more* minutes?" The answer is exactly the same as if you just walked up to the stop. The universe does not "owe" you a bus just because you waited. The past is deleted. Only the Exponential distribution has this property for continuous time.

Q3: Transforming Uniforms

Q3: Cutting the Stick

(b) **$\min(U, 1-U)$** *Analogy: The Folding Paper.* You have a stick of length 1. You break it at a random point U . You keep the **smaller** piece.

- Can the smaller piece be 0.9? No. The max size is 0.5 (breaking it exactly in the middle).
- If you break at 0.1, size is 0.1.
- If you break at 0.9, size is 0.1.
- Two outcomes map to the same size!
- This "folds" the probability density. The range shrinks to $(0, 0.5)$, but the density doubles to compensate. It becomes Uniform on $(0, 0.5)$.

Q6: Chip Battle

Q6: Wild vs Steady

The Concept: Variance is Risk.

- **Chip 2 (Steady):** Mean 22k, Low Variance. It is very reliable. It will almost always last around 22k hours.
- **Chip 1 (Wild):** Mean 20k, High Variance. It is erratic. Sometimes it dies early, sometimes it lasts forever.

The Battle:

- **Target 20k:** Steady Chip wins. It is safely above 20k. Wild Chip fails half the time.
- **Target 24k:** This is a very high bar. Steady Chip **loses**. It is *too* consistent; it never deviates enough to hit 24k. Wild Chip is the only one erratic enough to get lucky and hit 24k.
- *Moral:* If you need a miracle (very high target), pick the high-variance option.

6 Tutorial 6: Joint Distributions (Two Monkeys)

Q2: Variance of Sums

Concept Guide: Correlated Variance

What happens to the noise if you add two signals?

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$$

Analogy:

- **Uncorrelated:** Like two drunk people stumbling independently. They might cancel each other out. Variance adds normally.
- **Positively Correlated:** Like two people three-legged racing. They stumble together in the same direction. The variance **explodes** (extra 2Cov term).

In Q2, we sum n correlated variables. We have to sum the "Self-Variance" (Diagonal of grid) and the "Interaction Variance" (Neighbors in grid).

Q3: Central Limit Theorem (CLT)

Concept Guide: CLT: The Law of Averages

The Theorem: If you take an average of a massive group of monkeys, the result is always a **Gaussian (Bell) Curve**, even if the monkeys themselves are weirdly shaped. **The Benefit:** The Average is much more stable than the individual. **The Math:** If individual variance is σ^2 , the variance of the average of n items is σ^2/n . *Analogy:* One person's height is random. The average height of 100 people is very predictable.

Q5: Complex Random Variables

Q5: Rotating Noise

Analogy: The Spinning Plate. We have noise that is "Circularly Symmetric". Think of a round dinner plate centered at $(0,0)$.

- **Expectation:** The balance point is the center (0) . If you rotate the plate, the center is still (0) .
- **Variance:** The "spread" or radius of the plate. If you rotate the plate, it doesn't get bigger or smaller. Variance is constant.
- **Independence:** For a circular Gaussian, the "North-South" distribution is independent of the "East-West" distribution. Rotating it doesn't change this property.

7 Exam Paper: 2021 Semester 1 (2021S1)

Section A: Basic Applications

Question 1: Parking Slots (Probability)

Q1: The Monkey Analogy

The Setup: Imagine there are 10 chairs in a row. 6 monkeys come in and sit down in random chairs. This leaves 4 empty chairs. We want to know the chance that the 4 empty chairs are all sitting right next to each other (grouped together), rather than being scattered around the room.

Step-by-Step Solution:

1. **Total Combinations:** We have 10 slots and choose 4 to be empty.

$$\text{Total Ways} = \binom{10}{4} = \frac{10!}{4!(10-4)!} = 210$$

2. **Favorable Ways:** We want the 4 empty slots to be a single block (EEEE). Imagine the 10 slots are numbered 1 to 10. The group can start at position 1, 2, 3, 4, 5, 6, or 7. That is 7 possible "winning" arrangements.

3. **Probability:**

$$P = \frac{7}{210} = \frac{1}{30}$$

Question 2: Radar Power (PDF)

Q2: The Strong Monkey

The Setup: A monkey throws bananas at a wall. The strength of the throw follows an Exponential Distribution. A throw counts as a "hit" if it is stronger than the average.

Step-by-Step Solution:

1. **Identify Expectation:** The PDF is $f_Y(y) = \frac{1}{Q}e^{-y/Q}$. For this form, Mean $E[Y] = Q$.

2. **The Condition:** We need $P(Y > Q)$.

3. **Integrate:**

$$\begin{aligned} P(Y > Q) &= \int_Q^\infty \frac{1}{Q}e^{-y/Q} dy = \left[-e^{-y/Q} \right]_Q^\infty \\ &= 0 - (-e^{-1}) = e^{-1} \approx 0.368 \end{aligned}$$

Question 3: Bus Arrival (Poisson)

Q3: Waiting for the Truck

The Setup: The number of trucks (N) in t minutes is Poisson with mean $\lambda = t$. We want 80% certainty that at least one truck arrives.

Step-by-Step Solution:

1. **Lazy Monkey Strategy:** $P(N > 0) = 1 - P(N = 0)$.
2. **Set up:** $1 - P(N = 0) = 0.8 \Rightarrow P(N = 0) = 0.2$.
3. **Poisson Formula:** $P(0) = e^{-\lambda}$. Since $\lambda = t$, we have $e^{-t} = 0.2$.
4. **Solve:** $t = -\ln(0.2) \approx 1.61$ minutes.

Question 4: Mobile Plan (Geometric)

Q4: The Banana Subscription

(a) **Find p :** Geometric mean is $1/p$. Given Mean = 30.

$$30 = \frac{1}{p} \Rightarrow p = \frac{1}{30}$$

(b) **Cost PMF:**

- **Case 1 (Base Price):** If $M \leq 30$, Cost = \$20. $P(C = 20) = P(M \leq 30) = 1 - (1 - p)^{30} = 1 - \left(\frac{29}{30}\right)^{30}$.
- **Case 2 (Overcharge):** If $M = 30 + k$ (where $k \geq 1$), Cost = $20 + 0.5k$. $P(C = 20 + 0.5k) = P(M = 30 + k) = (1 - p)^{29+k} p = \left(\frac{29}{30}\right)^{29+k} \left(\frac{1}{30}\right)$.

Section B: Hypothesis Testing

Question 5: Battery Testing

Q5: The Lying Seller

The Setup: Seller claims $p \geq 0.8$. We suspect $p < 0.8$. Sample $n = 30$. We want Significance Level $\alpha \approx 5\%$.

(a) **Rejection Region:** We look at the lower tail (Cumulative Probability).

- $P(X \leq 20) \approx 6.1\%$ (Too high).
- $P(X \leq 19) \approx 2.5\%$ (Safest).

Decision: Reject Null Hypothesis if $X \leq 19$.

(b) **Decisions:**

- $X = 18$: Reject (18 is in region).
- $X = 19$: Reject (19 is in region).
- $X = 20$: Do Not Reject.

(c) **CLT Approximation:** $n = 100,000$, observed 80,000. Expected mean $np = 80,000$.

$$Z = \frac{\text{Observed} - \text{Mean}}{\text{Std Dev}} = \frac{80,000 - 80,000}{\sigma} = 0$$

$$\Phi(0) = 0.5$$

Question 6: Student Grades (Gaussian)

Q6: Exam Scores

(a) **Average Y :** Sum of independent Gaussians is Gaussian. Mean = 82. Variance of average = $\frac{\sigma^2}{n}$? No, $Y = 0.5X_1 + 0.5X_2$. $\text{Var}(Y) = 0.25(256) + 0.25(256) = 128$. $Y \sim N(82, 128)$.

(b) **Prob $Y \geq 90$:**

$$Z = \frac{90 - 82}{\sqrt{128}} \approx 0.707$$

$$P(Z \geq 0.707) = 1 - 0.7602 = 0.2398.$$

(c) **Max Score:** Easier to calculate "Both fail to get 90". Prob one exam < 90 : $Z = \frac{90-82}{\sqrt{16}} = 0.5$. $\Phi(0.5) = 0.6915$. Prob both < 90 : $0.6915^2 \approx 0.478$. Prob at least one ≥ 90 : $1 - 0.478 = 0.522$.

Question 7: Microelectronics (CLT)**Q7: The Chip Factory**

The Setup: Yield rate 0.9. Need 1000 good chips with 90% certainty. $P(K \geq 1000) \geq 0.9$.
Using Normal Approximation: Mean $0.9n$, SD $0.3\sqrt{n}$.

$$Z = \frac{1000 - 0.9n}{0.3\sqrt{n}} = -1.28$$

Solving quadratic for $\sqrt{n}$: $n \approx 1125$ chips.

8 Exam Paper: 2021/2022 Semester 1 (2122)

Part 1: Discrete Random Variables

Question 1: Probabilities vs. Averages

Q1: The Banana Box

Goal: Find $P(X < E[X])$.

(a) **Binomial** ($n = 6, p = 0.5$) Mean $E[X] = 3$. We want $P(X < 3) = P(0) + P(1) + P(2)$. Sum = $0.0156 + 0.0938 + 0.2344 = 0.34375$.

(b) **Geometric** ($p = 1/3$) Mean $E[X] = 1/p = 3$. We want $P(X < 3) = P(1) + P(2)$. $P(1) = 1/3$. $P(2) = (2/3)(1/3) = 2/9$. Sum = $5/9$.

(c) **Poisson** ($\lambda = 3$) Mean $E[X] = 3$. We want $P(X < 3) = P(0) + P(1) + P(2)$. $e^{-3}(1+3+4.5) = 8.5e^{-3} \approx 0.423$.

Question 2: Function Analysis

Q2: Disguised Binomial

Given $P(x) = \binom{5}{x} 0.3^{5-x} 0.7^x$. This is Binomial with $n = 5, p = 0.7$.

(a) **Standard Deviation:** $Var = np(1-p) = 5(0.7)(0.3) = 1.05$. $SD = \sqrt{1.05} \approx 1.025$.

(b) $E[X^2 - 1]$: Use $Var = E[X^2] - (E[X])^2$. $E[X] = 3.5$. $E[X^2] = 1.05 + 3.5^2 = 13.3$. Answer: $13.3 - 1 = 12.3$.

Part 2: Continuous Random Variables

Question 3: Validating a PDF

Q3: The Smooth Banana

$f(x) = ax^2 + bx$ on $[0, 1]$.

(a) **Conditions:** Area must be 1: $\int_0^1 (ax^2 + bx) dx = a/3 + b/2 = 1 \Rightarrow b = 2 - 2a/3$. Non-negative: $f(x) \geq 0$ on $[0, 1]$. This restricts a to approximately $[-6, 3]$.

(b) **Find b if a=3:** $3/3 + b/2 = 1 \Rightarrow 1 + b/2 = 1 \Rightarrow b = 0$.

Question 4: Uniform Variable

Q4: Cutting the Stick

Z is Uniform on $[-k, k]$ (since mean is 0). Variance $\sigma^2 = k^2/3$. **Goal:** $P(Z^2 < \sigma^2)$.

$$P(Z^2 < k^2/3) = P(-\frac{k}{\sqrt{3}} < Z < \frac{k}{\sqrt{3}})$$

Since distribution is flat, prob is ratio of lengths:

$$\frac{2k/\sqrt{3}}{2k} = \frac{1}{\sqrt{3}} \approx 0.577$$

Part 3: Applied Problems

Question 5: The Jogger

Q5: Motivation U

$X = 6 + U^3$ where $U \sim \text{Uniform}(-1, 1)$.

(a) **E[X]:** $E[U^3] = 0$ (Symmetry). $E[X] = 6$.

(b) **Var(X):** $E[X^2] = E[(6 + U^3)^2] = 36 + 12E[U^3] + E[U^6]$. $E[U^6] = \int_{-1}^1 u^6 (1/2) du = 1/7$.
 $\text{Var}(X) = (36 + 1/7) - 36 = 1/7$.

(e) **P(Z ≥ 255):** Z is sum of 42 kms. $E[Z] = 42(6) = 252$. $\text{Var}(Z) = 42(1/7) = 6$. $SD = \sqrt{6} \approx 2.45$.

$$Z_{\text{score}} = \frac{255 - 252}{2.45} = 1.22$$

$$P(Z_{\text{score}} \geq 1.22) = 1 - 0.8888 = 0.1112.$$

Question 6: Train vs Taxi

Q6: The Exponential Race

Train $\lambda_X = 1/10$. Taxi $\lambda_Y = 1/5$.

(a) **Prob Train First:**

$$\frac{\lambda_X}{\lambda_X + \lambda_Y} = \frac{0.1}{0.3} = 1/3$$

(d) **The Regret:** You see Train (wait 0), total time 30. You reject it. Now you need Taxi to beat 30 mins. Taxi takes 25 mins ride. So Taxi must arrive in < 5 mins. $P(Y < 5) = 1 - e^{-0.2(5)} = 1 - e^{-1} \approx 0.632$.

9 Exam Paper: 2023/2024 Semester 1 (2324)

Section A: Discrete Probabilities

Question 1: The Dice Roll

Q1: Rolling for Success

Event A: Roll > 3 (4, 5, 6). $p = 0.5$.

(a) **Binomial:** 3 wins in 10 rolls. $\binom{10}{3}(0.5)^{10} \approx 0.117$.

(b) **Negative Binomial:** 5th roll is 3rd win. Needs 2 wins in first 4 rolls, then Win. $\binom{4}{2}(0.5)^4 \times 0.5 \approx 0.1875$.

(c) **Geometric:** At least 4 rolls for 1st win. Means 3 failures first. $(0.5)^3 = 0.125$.

(d) **Race:** A happens before '1'. $P(A) = 3/6$. $P(1) = 1/6$.

$$\frac{P(A)}{P(A) + P(1)} = \frac{3}{4} = 0.75$$

Question 2: Best-of-5 Game

Q2: Tennis Monkeys

First to 3 wins. $p = 0.5$.

(a) **Expectation:** Possible lengths: 3, 4, 5. $P(3) = 2(0.5)^3 = 0.25$. $P(4) = \binom{3}{2}(0.5)^4 \times 2 = 0.375$. $P(5) = 0.375$. $E[X] = 4.125$ rounds.

(b) **Conditional:** Given A wins R1. A needs 2 wins before B gets 3. $P = 0.6875$.

Section A: Continuous Random Variables

Question 3: Triangle Distribution

Q3: The Triangle

PDF is triangle on $[0, 2]$ peak at 1.

(b) **Mean/Var:** Symmetric at 1, so Mean = 1. Variance Calculation gives $1/6$.

(c) **Transformation:** $Y = 1/X$. $F_Y(y) = P(1/X \leq y) = P(X \geq 1/y) = 1 - F_X(1/y)$.

Question 4: Joint Probability

Q4: The Cloud

$f(x, y) = c(x + y)$ on $x \in [0, 2], y \in [0, 1]$.

(a) **Find c:** $\int \int c(x + y) = 1 \Rightarrow 3c = 1 \Rightarrow c = 1/3$.

(c) **Covariance:** $Cov = E[XY] - E[X]E[Y] = -1/81$. Small negative correlation.

Section B: Applied Problems

Question 5: Shooting Game

Q5: Turn-Based Combat

$P_A = 0.4, P_B = 0.6$. A shoots first.

(a) **A wins on 2nd shot:** A Miss, B Miss, A Hit. $0.6 \times 0.4 \times 0.4 = 0.096$.

(b) **A wins eventually:** Geometric Series Sum. $P = \frac{0.4}{1 - (0.6)(0.4)} = \frac{0.4}{0.76} \approx 0.526$.

Question 6: Round-off Error**Q6: CLT on Errors**

Error Δ is Uniform $[-0.5, 0.5]$. Mean 0, Var $1/12$. Sum of 50 errors: Mean 0, Var $50/12 \approx 4.16$.
 $P(|S| > 3)$: $Z = 3/\sqrt{4.16} \approx 1.47$. Using Q-function: $2 \times Q(1.47) \approx 0.14$.

10 Exam Paper: 2024/2025 Semester 1 (2425)

Section A: Discrete Probabilities

Question 1-1: The Biased Coin

Q1-1: Finding p

Given $P(HH) = 0.36 \Rightarrow p^2 = 0.36 \Rightarrow p = 0.6$.

(a) **Two Tails:** $(1 - 0.6)^2 = 0.16$.

(b) **At least 5 flips:** Means 4 failures first. $(0.4)^4 = 0.0256$.

(c) **Conditional:** Given 3 Heads in 10. Prob 5th is the 1st Head. Means T T T T H and then 2 Heads in last 5. Numerator: $\binom{5}{2}$. Denominator: $\binom{10}{3}$. $10/120 = 0.0833$.

Question 1-2: Set Theory

Q1-2: Overlaps

Using $P(B|A) = P(AB)/P(A)$. Let $x = P(AB)$. $P(A) = x + 0.15$ (Overlap + Non-Overlap). $0.19 = x/(x + 0.15)$. Solving gives $x \approx 0.035$. Total $P(A) \approx 0.185$.

Question 2: Ten-Sided Die

Q2: Weird Die

Sides: 1, 2, 2, 3, 3, 3, 4, 4, 4, 4. $P(1) = 0.1, P(2) = 0.2, P(3) = 0.3, P(4) = 0.4$.

(a) **Independence:** $P(X_1 \leq 3) \times P(X_2 > 2) = 0.6 \times 0.7 = 0.42$.

(b) **Expectation/Var:** $E[X] = 3$. $Var[X] = 1$. $Var(Sum) = 1 + 1 = 2$.

Section A: CLT

Question 3: The Big Coin Flip

Q3: CLT with Continuity

$n = 400, p = 0.2$. Mean 80, Var 64, SD 8.

(b) **Prob** $63 < X \leq 88$: Continuity Correction: $[63.5, 88.5]$. $Z_1 = (63.5 - 80)/8 \approx -2.06$. $Z_2 = (88.5 - 80)/8 \approx 1.06$. Area = $Q(-2.06) - Q(1.06)$.

Section B: Continuous Variables

Question 4: Piecewise Function

Q4: The Split Curve

PDF: ax^2 on $[-2, 0]$ and bx on $[0, 1]$. Two unknowns, two eqns (Area=1, Mean=-5/12). Solve: $a = 3/16, b = 1$.

(c) **Transformation:** $Y = 1/|X|$. Maps $[-2, 1]$ to $[0.5, \infty)$.

Question 5: Joint Distribution

Q5: The Triangle Cloud

Region $0 < y < 1 - x$. (a) **Find c:** Integral gives $c = 24$. (c) **Conditional:** Requires careful integration over the region where $Y < 0.5$ and $X > Y$. Result: $8/11 \approx 0.727$.

Found: <https://github.com/brianstm/NUS.git>